

RA'ED ALASMAR

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PROFESSIONAL SUMMARY

Electrical Engineering student specializing in intelligent communication systems, with hands-on RAN experience through internships at **Zain Jordan** on **Nokia** equipment. Completed the full **CCNA** track. Now researching AI-driven RAN optimization at **Nokia Bell Labs**, alongside AI-based anomaly detection research and leading a RAG faculty chatbot with **GDG on Campus BME**.

WORK EXPERIENCE

AI Research Collaborator - Nokia Bell Labs

05/2026 – present

- Researching AI-based methods for autonomous optimization of 5G networks.
- Developing and benchmarking multi-agent reinforcement learning for self-organizing network (SON) functions against rule-based methods.

Tech Team Member, Project Lead - Google Developer Group on Campus BME

02/2026 – present

- Lead the AI Assistant project, a RAG-based AI chatbot serving students at BME's engineering faculty.
- Designed a multi-stage retrieval pipeline with Cohere reranking and Gemini 2.5 Flash.

C Programming Lab Instructor - Budapest Univ. of Tech. & Econ.

09/2025 - 01/2026

- Led weekly labs for ~25 students, teaching C from fundamentals to real-life projects.
- Mentored students through algorithms, data structures, and system-level problem solving.

RAN Optimization Intern, Zain Jordan

07/2025 - 09/2025

- Analyzed LTE/5G KPIs including handover success, call drops, RSRP/RSRQ/SINR, and throughput.
- Used OSS tools and RTWP/spectrum scans to detect uplink interference and optimize coverage.
- Supported frequency planning, BSIC checks, and handover optimization with cross-functional teams.

RAN Implementation Intern, Zain Jordan

06/2025 - 07/2025

- Assisted in full RAN site implementation: BTS setup, ERP ordering, and microwave link deployment.
- Validated BBU, RU, IDU/ODU, and RF jumpers; monitored KPIs and alarms via OSS systems.

PROJECTS & RESEARCH

Enterprise Office Building Infrastructure

- Designed a VLAN-segmented enterprise network with OSPF, DHCP, inter-VLAN routing, SSH, and MAC-based port security in **Cisco Packet Tracer**.

Software-Defined Network Traffic Guardian (Python/Linux)

- Built a Python SDN controller using Ryu to manage OpenFlow switches in Mininet, with automated real-time traffic rerouting to resolve congestion.

Task Manager (C++)

- Engineered a CLI task manager with dynamic memory management, task prioritization.

AI-Based Anomaly Detection Research

- Replaced majority voting with AutoML ensemble routing to select the best AI model.
- Optimized ensemble weights and detection thresholds to improve F1 score by ~14%.

EDUCATION

Bachelor of Electrical Engineering

09/2022 - 01/2027

Budapest University of Technology and Economics

- Received the Stipendium Hungaricum full scholarship award (top 10% acceptance rate).
- Member of BME Sollertia, bridging international and Hungarian students.

CERTIFICATES

- **Cisco Networking Academy** - CCNA (Full Track): 140+ labs, 50+ hours.
- **DeepLearning.AI** - Machine Learning Specialization.

SKILLS

Telecommunications: UMTS/LTE/NR Architecture, RAN Optimization, KPI Analysis, Nokia RAN Equipment, OSS Systems.

Networking: Router/Switch Configuration, OSPF, EIGRP, VLANs, DHCP, NAT, Subnetting, ACLs.

Programming: Python, C, C++, SQL.

AI & ML: RAG Frameworks, Anomaly Detection, AutoML (TPOT), Ensemble Methods, LLM Integration.

Tools: Git & GitHub, Jira, MS Office, Packet Tracer, Linux CLI.